BEFORE YOU READ

How smooth is the derivative?

Functions sort into classes by how well their derivatives behave. Each class sits inside the one around it.

$$C^\infty \subset C^1 \subset D \subset C^0$$

C^0 — Continuous

f' may not exist

`|x|`

D — Differentiable

f' exists everywhere but may be discontinuous

`x² sin(1/x), f(0) = 0`

C^1 — Continuously differentiable

f' exists and is continuous

`polynomials, sin x, eˣ`

C^∞ — Smooth

all higher derivatives exist and are continuous

TAKEAWAY

Differentiable is not the same as C^1

A derivative can exist everywhere and still be discontinuous, as $x^2 \sin(1/x)$ shows; by Darboux's theorem it can never have a jump.